ALEPH Charged-Particle Cumulants

Status Report

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Current Status

- Standalone ROOT/C++ cumulant analysis using charged particles in archived ALEPH LEP1 data.
- Current data input is the merged LEP1 1992–1995 StudyMult sample with **3,050,610 events**.
- Main data/MC results are detector-level; MC-only generator-level reference plots are included.
- Outputs include inclusive, MC-only, generator-level MC-only, rapidity-gap, two-subevent, MC-comparison, and generator-level WW signed-shoving plots.

Important caveat: no converted 1991 ROOT sample was found in the checked merged StudyMult directories, so the current data result is 1992–1995 rather than complete 1991–1995 LEP1.

Samples and Selection

Data

- `LEP1Merged_20200611.root`
- Tree: `t`, StudyMult fixed-array format
- 1992: 551,474 events
- 1993: 538,601 events
- 1994: 1,365,440 events
- 1995: 595,095 events

MC samples

- Reco: `LEP1MCMerged.root:t`, 771,597 entries
- Gen reference: `LEP1MCMerged.root:tgen`, 771,597 entries
- Reco MC is the 1994 baseline comparison

Track/event settings

- StudyMult charged flags: `pwflag = 0,1,2`
- $p_T > 0.4$ GeV; thrust plots use thrust-axis p_T
- multiplicity bins: $[0, 10), \ldots, [40, \infty)$

Analysis Workflow

- Q-cumulants are accumulated for harmonic $n = 2$: $\langle 2 \rangle$, $\langle 4 \rangle$, $\langle 6 \rangle$, $\langle 8 \rangle$.
- Derived observables: $v_2\{2\}$, $v_2\{4\}$, $v_2\{6\}$, $v_2\{8\}$.
- Each production is split into 16 chunks; statistical errors use delete-one-chunk jackknife after recomputing the nonlinear cumulant values.
- Internal self-test compares ordered-correlator implementation with brute-force distinct-particle sums.
- Comparison plots use 1992–1995 data and the 1994 reconstructed-MC baseline; MC-only reco and gen-level plots are shown separately.

Cumulant Definitions

- Within each $N_{trk}^{offline}$ bin, event sums are combined into weighted correlators before forming cumulants:

$$\langle\langle 2k\rangle\rangle = \frac{\sum_{\text{events}} \mathrm{Num}_{2k}}{\sum_{\text{events}} \mathrm{Den}_{2k}}, \qquad \mathrm{Num}_{2k} = \mathrm{Re} \sum_{\text{distinct}} e^{2i(\phi_1+\cdots+\phi_k-\phi_{k+1}-\cdots-\phi_{2k})}.$$

$$\begin{aligned}
c_2\{2\} &= \langle\langle 2\rangle\rangle, \\
c_2\{4\} &= \langle\langle 4\rangle\rangle - 2\langle\langle 2\rangle\rangle^2, \\
c_2\{6\} &= \langle\langle 6\rangle\rangle - 9\langle\langle 4\rangle\rangle\langle\langle 2\rangle\rangle + 12\langle\langle 2\rangle\rangle^3, \\
c_2\{8\} &= \langle\langle 8\rangle\rangle - 16\langle\langle 6\rangle\rangle\langle\langle 2\rangle\rangle - 18\langle\langle 4\rangle\rangle^2 \\
&\quad + 144\langle\langle 4\rangle\rangle\langle\langle 2\rangle\rangle^2 - 144\langle\langle 2\rangle\rangle^4.
\end{aligned}$$

Den_{2k}: Inclusive denominators are $N(N-1)\cdots(N-2k+1)$

two-subevent denominators are $N_A(N_A-1)\cdots(N_A-k+1)\,N_B(N_B-1)\cdots(N_B-k+1)$.

Extracting $v_2\{2k\}$

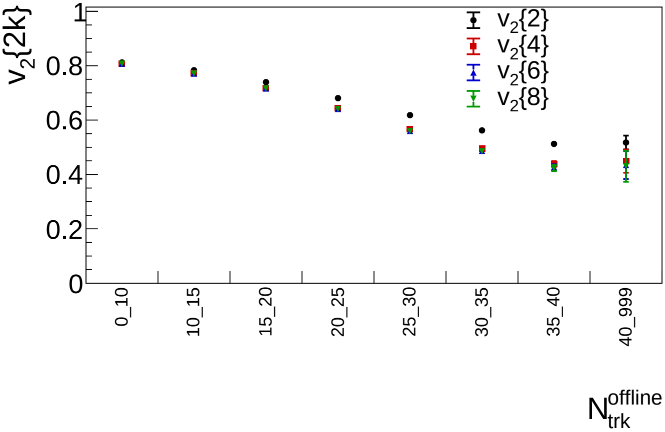
$$v_2\{2\} = \sqrt{c_2\{2\}},$$
$$v_2\{4\} = (-c_2\{4\})^{1/4},$$
$$v_2\{6\} = (c_2\{6\}/4)^{1/6},$$
$$v_2\{8\} = (-c_2\{8\}/33)^{1/8}.$$

Valid real roots require

$$c_2\{2\} \geq 0, \quad c_2\{4\} < 0,$$
$$c_2\{6\} > 0, \quad c_2\{8\} < 0.$$

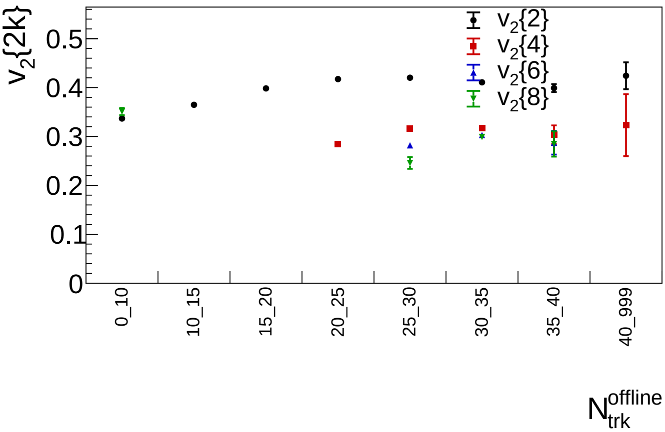
- The two-subevent version uses the same cumulant algebra, but each $\langle\langle 2k \rangle\rangle$ samples k tracks from $\eta > 0$ and k tracks from $\eta < 0$, where η is defined relative to the plotted axis.
- Statistical errors are evaluated with 16 delete-one-chunk jackknife samples after recomputing the nonlinear $c_2\{2k\} \to v_2\{2k\}$ transformation.
- A $v_2\{2k\}$ bin is not drawn if the central value or any jackknife sample fails the required real-root sign.

Inclusive Result: Beam Axis



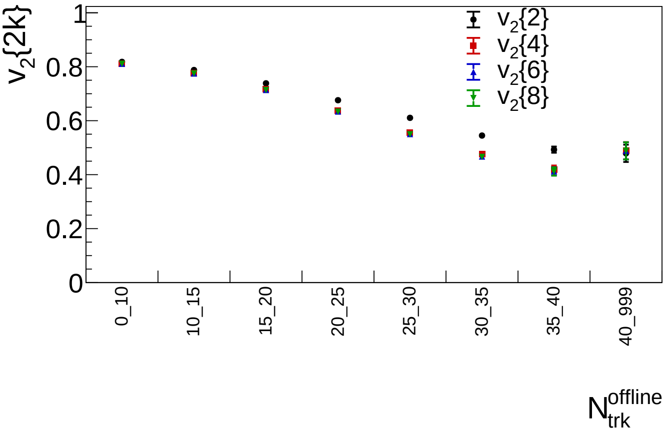
Merged LEP1 1992–1995 data; detector-level charged particles with $p_T > 0.4$ GeV.

Inclusive Result: Thrust Axis



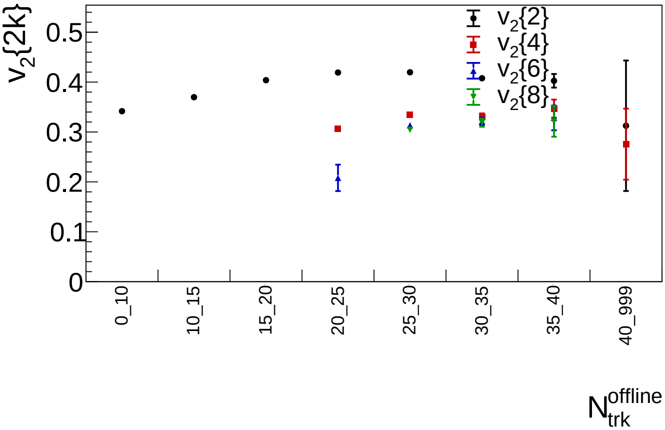
Same lab-frame selected charged-particle multiplicity bins, with azimuth evaluated around the event thrust axis.

MC-Only Inclusive: Beam Axis



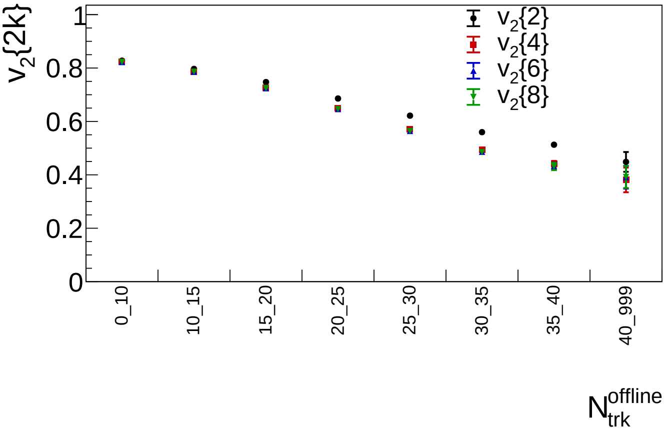
1994 reconstructed MC baseline; same charged-particle selection and lab multiplicity bins.

MC-Only Inclusive: Thrust Axis



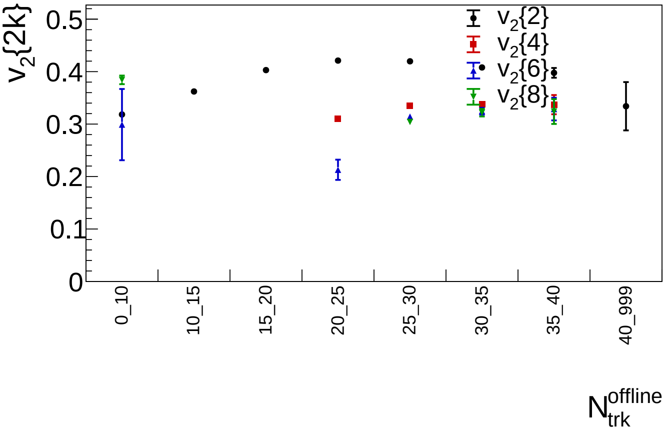
1994 reconstructed MC baseline with azimuth and pseudorapidity evaluated around the event thrust axis; bins use lab selected multiplicity.

MC-Only Gen Level: Beam Axis



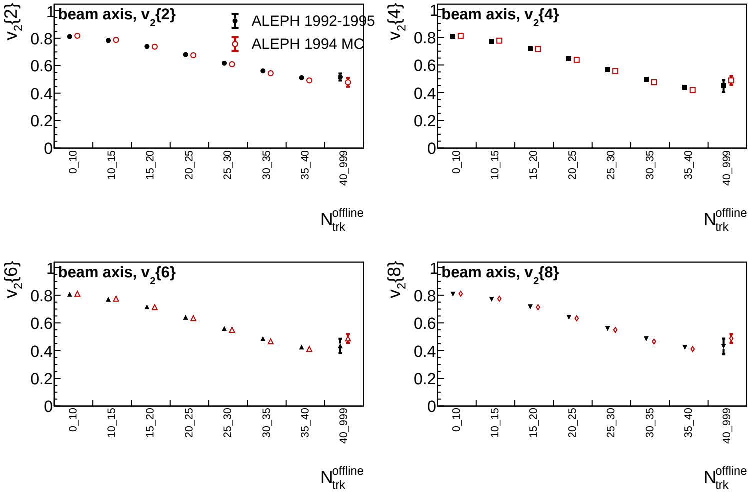
Generator-level charged particles from `LEP1MCMerged.root:tgen`; $p_T > 0.4$ GeV and the same multiplicity bin edges as detector-level plots.

MC-Only Gen Level: Thrust Axis



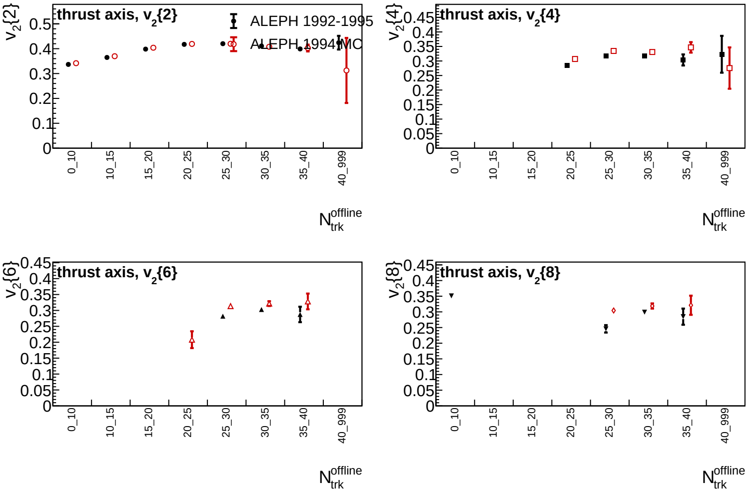
Generator-level `tgen` MC with azimuth and pseudorapidity evaluated around the event thrust axis; bins use selected gen-level charged multiplicity.

Data/MC Comparison: Beam Axis



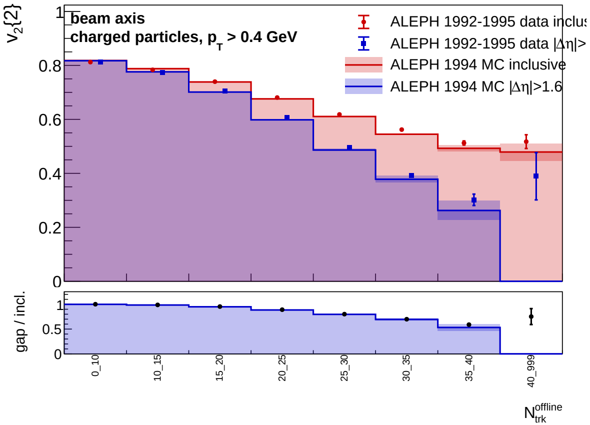
Data: LEP1 1992–1995; MC: 1994 reconstructed baseline.

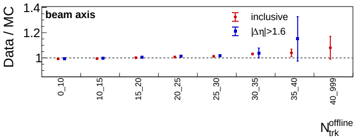
Data/MC Comparison: Thrust Axis



Data: LEP1 1992–1995; MC: 1994 reconstructed baseline.

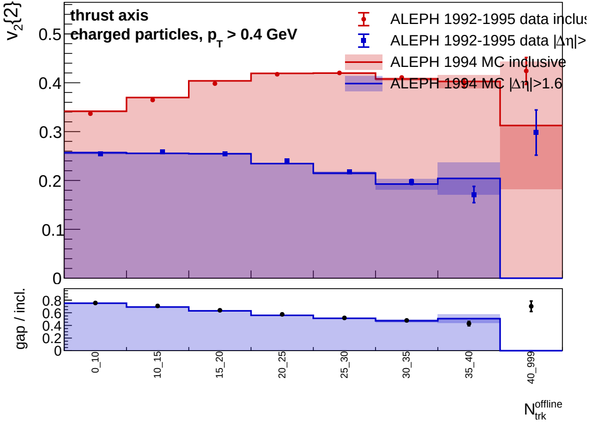
Rapidity Gap: $|\Delta\eta| > 1.6$, Beam Axis

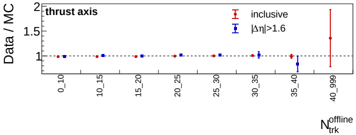




- Looser than the $|\Delta\eta| > 2.0$ study.
- Ratio panel tracks data/MC for inclusive and gap observables.
- Largest uncertainty remains in the high-multiplicity tail.

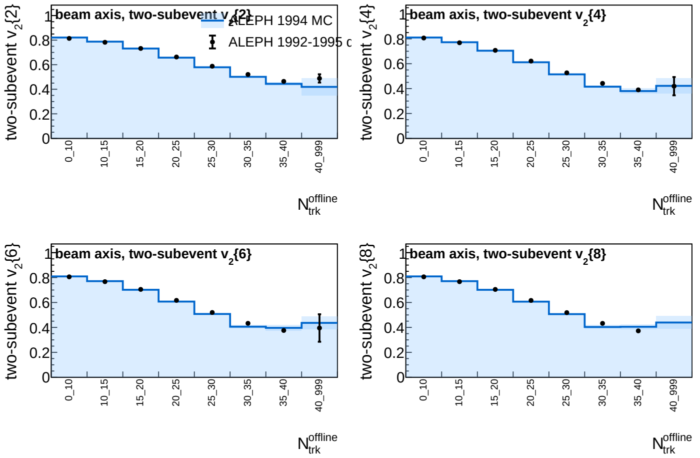
Rapidity Gap: $|\Delta\eta| > 1.6$, Thrust Axis





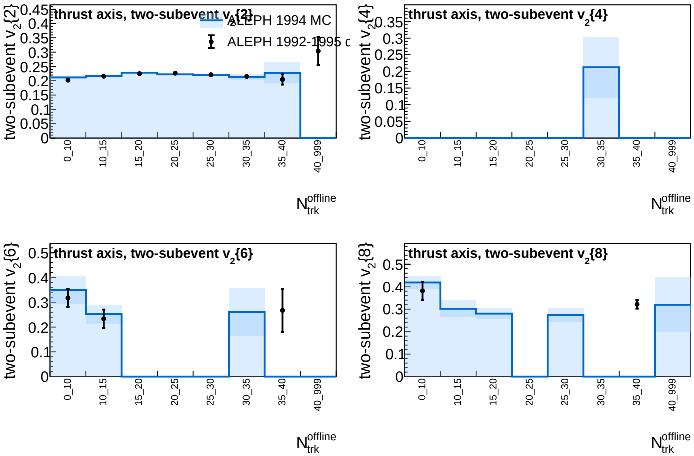
- Gap is evaluated in pseudorapidity around the thrust axis.
- Suppression is stronger than in the beam-axis view.
- Data and MC follow the same broad multiplicity trend.

Two-Subevent Cumulants: Beam Axis



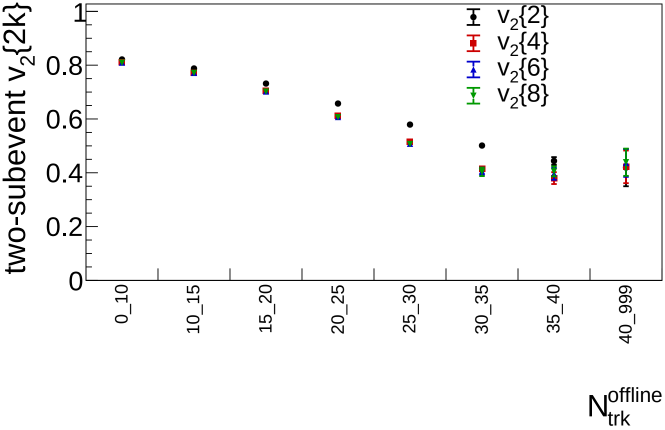
Two-subevent result samples k particles from beam-axis $\eta > 0$ and k particles from beam-axis $\eta < 0$ for $v_2\{2k\}$.

Two-Subevent Cumulants: Thrust Axis



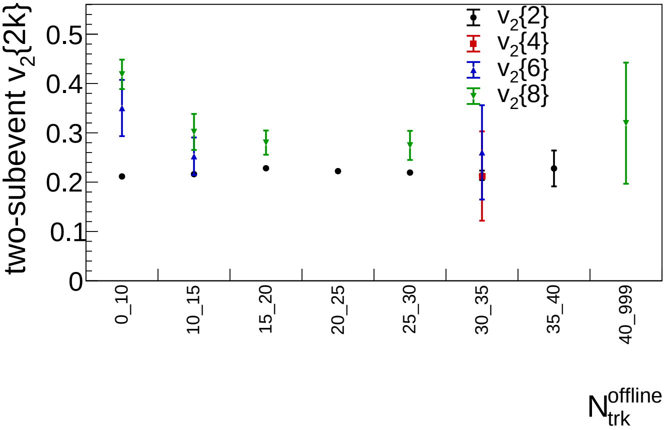
Two-subevent result samples k particles from thrust-axis $\eta > 0$ and k particles from thrust-axis $\eta < 0$ for $v_2\{2k\}$.

Two-Subevent MC Only: Beam Axis



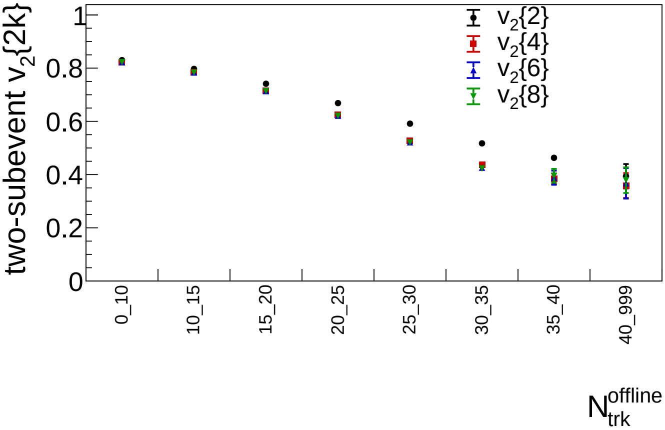
1994 reconstructed MC baseline; two-subevent cumulants only, using beam-axis $\eta > 0$ and $\eta < 0$ subevents.

Two-Subevent MC Only: Thrust Axis



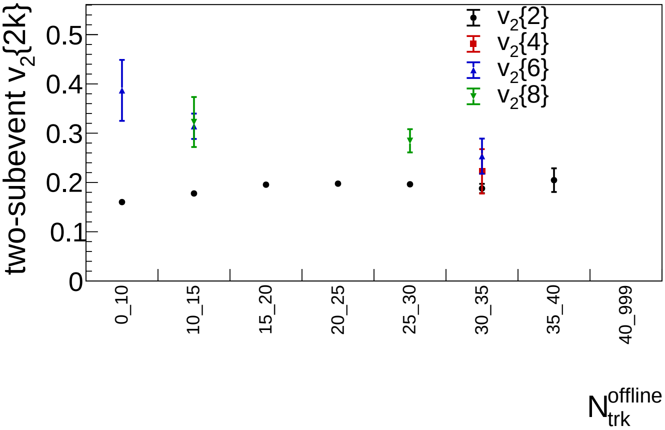
1994 reconstructed MC baseline; thrust-axis two-subevent cumulants use thrust-axis $\eta > 0$ and $\eta < 0$ subevents.

Two-Subevent MC Gen Level: Beam Axis



Generator-level `tgen` two-subevent cumulants, with k particles sampled from beam-axis $\eta > 0$ and k from $\eta < 0$.

Two-Subevent MC Gen Level: Thrust Axis



Generator-level `tgen` two-subevent cumulants with the event thrust axis defining $\eta > 0$ and $\eta < 0$ subevents.

WW200 Signed Shoving Test: Scope

Samples used

- Generator-level hadronic $e^+e^- \to W^+W^- \to q\bar{q}q\bar{q}$ at $\sqrt{s} = 200$ GeV.
- Three already-produced 5M-event trees: no shoving, nominal Gleipnir shoving, enhanced shoving.
- Charged final-state particles; $p_T > 0.4$ GeV; same $N_{trk}^{offline}$ bins as the cumulant analysis.
- No new event generation was done for this signed test.

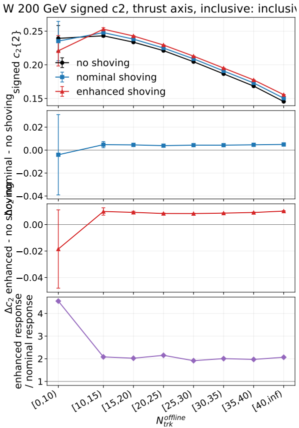
Inventory outcome

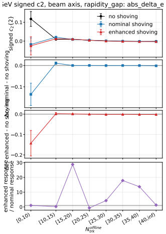
- Available: final-particle momenta, PID, charged flags, charge, event-selection flag, final-particle vertices.
- Not available: full Pythia event record, W-daughter links, mother/daughter indices, parton records, string pieces, Gleipnir space-time records.
- Therefore this is a signed two-particle response test, not a direct geometry-plane sign measurement.

Signed Observables and Uncertainties

- The usual $v_2\{2\} = \sqrt{c_2\{2\}}$ is positive when $c_2\{2\} > 0$, so it cannot test the response sign.
- Signed pair correlator:
$$c_2\{2\} = \langle \cos 2(\phi_i - \phi_j) \rangle \,, \qquad \Delta c_2 = c_2^{\text{shoving}} - c_2^{\text{no shoving}}.$$
- Computed for inclusive pairs, $|\Delta\eta| > 1.0, 1.6, 2.0, 2.5, 3.0$, and two-subevent pairs $\eta < -b$ vs. $\eta > b$ with $b = 0, 0.5, 0.8, 1.0$.
- Long-range thrust-axis coefficients use $1.6 < |\Delta\eta| < 3.2$: $V_{n\Delta} = \langle \cos n\Delta\phi \rangle$, with $v_n^{\text{sgn}} = \text{sign}(V_{n\Delta})\sqrt{|V_{n\Delta}|}$.
- Statistical errors use the 16-block delete-one-block jackknife; sample differences use $\sqrt{\sigma^2_{\text{shoving}} + \sigma^2_{\text{no}}}$.

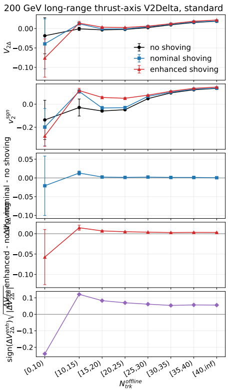
Signed $c_2\{2\}$: Shoving Response

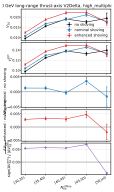




- Inclusive thrust-axis $c_2\{2\}$ increases with shoving.
- For $30 \leq N_{trk}^{offline} < 35$: $\Delta c_2 = +0.00430$ nominal and $+0.00862$ enhanced.
- Beam-axis $|\Delta\eta| > 1.6$ can have negative enhanced response; same bin gives -9.18×10^{-4}, about -2.34σ.

Long-Range Thrust-Axis $V_{2\Delta}$





- Pair window: $1.6 < |\Delta\eta| < 3.2$ in thrust-axis coordinates.
- Standard bins show mostly positive $\Delta V_{2\Delta}$.
- In $[30, 35)$, $V_{2\Delta}$ is 0.00953, 0.01078, 0.01244 for no, nominal, enhanced shoving.
- Enhanced response is usually larger than nominal; the high-multiplicity tail is less stable.

Signed Study Interpretation

- The signed observables show a statistically significant generator-level shoving response in several Δc_2 and $\Delta V_{2\Delta}$ bins.
- The response sign is not universal: it depends on the axis definition and on the nonflow-suppression handle.
- The thrust-axis long-range $V_{2\Delta}$ response is mostly positive; the beam-axis rapidity-gap response can become negative in mid/high multiplicity bins.
- The stored WW trees do not support a clean geometry-plane signed v_2^{geom} test, because partons, strings, and truth daughter links are absent.
- This is generator-level only; it is not a claim of against-flow in WW data.

What We Learn So Far

- Beam-axis $v_2\{m\}$ values are large and decrease with multiplicity; high-order cumulants are stable in the populated bins.
- Thrust-axis values are smaller and high-order cumulants become sign-invalid in several low-multiplicity bins.
- Rapidity-gap selections reduce the inclusive two-particle coefficient, consistent with sizable short-range or jet-like contributions.
- The 1994 MC baseline broadly follows the data trends, but this is not yet a detector-corrected or systematic-uncertainty result.

Next Steps

- Add correction and systematic studies: tracking, event selection, axis definition, MC model dependence, and multiplicity migration.
- Stress-test rapidity-gap and two-subevent definitions against alternate split/gap choices.
- Expand MC comparisons beyond the current 1994 baseline where available.
- Keep the analysis note and reproducible run instructions synchronized with each production.